

Fanny Lovisa Persson, M.Sc.

Molecular Biologist | fanny.p98@me.com | [LinkedIn](#) | +46 72 710 99 82

Meticulous and motivated molecular biologist currently working as a research assistant in viral disease surveillance, focused on method development. Passionate about advancing knowledge in virology and immunology, particularly in the context of host-pathogen interactions and therapeutic innovation. Diligent and hardworking. Driven to solve complex problems within passionate high-performance teams. High EQ, great communicator, people-person. Technically competent and experienced within various academic and commercial-grade laboratory environments and standards (GDP/GMP).

EDUCATION

AUGUST 2022 — JUNE 2024	Master of Science: Microbiology and Immunology, Uppsala University, Sweden Research-oriented training in molecular microbiology, immunology and infection biology, with strong emphasis on experimental design, data analysis, and laboratory techniques. Graduated with a 4.5 out of 5 average grade.
AUGUST 2023 — DECEMBER 2023	Exchange: University of Edinburgh, Scotland Expanded my scientific knowledge in immunology, genetic medicine and molecular biology with a particular interest in infection biology. I received grade B in all courses.
AUGUST 2019 — JUNE 2022	Bachelor of Science: Cell and molecular biology, Minor in Chemistry, Uppsala University, Sweden Built a strong academic foundation in the study of cell and molecular biology, acquired practical skills through laboratory work, and developed confidence in presenting scientific results.

RESEARCH EXPERIENCE

SEPTEMBER 2023 — JUNE 2024	M.Sc. thesis project, Uppsala University, Sweden Investigated the presence and load of respiratory viruses (SARS-CoV-2, Influenza A & B, RSV) in Swedish wastewater, contributing to the development of wastewater-based epidemiology (WBE) as a public health surveillance tool in Sweden. Utilized RT-qPCR, nucleic acid extraction techniques, R language, qPCR-assay development tools. Collaborated with the Swedish Public health agency and Umeå University researchers. I produced novel results contributing to the area of research.
JANUARY 2023 — JUNE 2023	B.Sc. thesis project, Uppsala University, Sweden In my project titled “Deficiencies in the delta subunit of the CD3 protein causes severe combined immunodeficiency” I explored T-lymphocyte development and how an impairment in this process can cause the syndrome severe combined immunodeficiency. I produced a project report in the format of a literature review.

WORK EXPERIENCE

MAY 2024 — PRESENT	Research assistant: Uppsala University and Swedish University of Agricultural Sciences, Sweden After finishing my master's thesis under the supervision of Associate Professor Anna J. Székely, I earned a research assistant role in her group. Main tasks include weekly sample analysis for the presence of five viral targets in wastewater from 19 cities in Sweden, method development and implementing new assays to cover more targets in our weekly surveillance. Spearheaded a new project on surveillance of avian influenza, typing of influenza A and expanding the surveillance to cover bacterial pathogens. I also consult other research groups in the division on specifically qPCR and sequencing for their unique targets, including bacteriophages and bacterial DNA and RNA.
APRIL 2021 — FEBRUARY 2024	Pharmacology and therapeutics technician: Svensk Dos AB, Uppsala At Svensk Dos, a major Swedish pharmaceutical therapeutics company, I ensured compliance to GMP and GDP in commercial pharmaceutical clean rooms. Regularly commended for my proficiency and attention to detail. Executed various tasks in pharmaceutical production, emphasizing fast, efficient, and precise work for patient safety.
APRIL 2023 — SEPTEMBER 2023	Field assistant in biological mosquito control: Uppsala University in collaboration with municipality of Nedre Dalälven Gained experience in experimental assay development as part of a research group establishing viral presences in Swedish mosquito species and effectively controlling mosquito populations. I worked in the field capturing specific species and prepared assays for viral detection as well as sampled mosquito populations in areas affected by flooding. I honed crucial skills such as effective communication, efficiency, stress tolerance, and working independently.

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PRESENTATIONS AND MEDIA APPEARANCE

CONFERENCE TALK	Persson, F. The use of wastewater monitoring in epidemiology to track and profile influenza virus A, influenza virus B, RSV and SARS-CoV-2. 2024. Conference: Infectious disease epidemiology: surveillance, forecasting and modelling, Stockholm, Sweden
MEDIA APPEARANCE	Participated in a video segment and article on wastewater-based surveillance of respiratory viruses in Sweden. Successfully explaining to a broad range of viewers how this disease monitoring is done. Channel: SVT, Swedish national news
MEDIA APPEARANCE	Spoke on the radio about wastewater-based surveillance of respiratory viruses, specifically addressing a dramatic increase in measured Influenza A. Channel: Uppland radio P4, Sweden

SKILLS AND ADDITIONAL INFORMATION

LABORATORY	– Nucleic acid extraction using both bead-based and silica column-based methods, including from complex environmental samples (e.g., wastewater). – RT-qPCR, qPCR, endpoint PCR, cDNA synthesis, and RNA purification. – Bacterial cultivation, gene cloning into fusion vectors, and transformation protocols. – Protein and nucleic acid analysis techniques: SDS-PAGE, agarose gel electrophoresis, Western blot, Northern blot, ELISA, EMSA, size-exclusion chromatography. – Bacterial motility assays and functional studies on virulence factors (gain- and loss-of-function approaches). – Basic animal handling experience, including dissection and administration of test compounds to rats.
BEST PRACTICE	– Experience working in research laboratory (>1 year) – Experience working in production cleanrooms (>1 year) – Good Manufacturing Practice (GMP) – Good Distribution Practice (GDP)
DIGITAL	– Microsoft Suite – Coding languages: R-studio and Python – Clustermarket – LabCollector Electronic Lab Notebook
LANGUAGES	Swedish (native), English (fluent)
HOBBIES	In my spare time I am a dedicated musician involved in bands and performing acts. I am a massive fan of traditional music and blues. I am an avid skier and snowboarder, and I love learning new skills and trying something new.

REFERENCES

Anna J. Székely, M.Sc., PhD	Associate professor and researcher leading the Swedish environmental epidemiology center at SLU Anna.szekely@slu.se +46 73 974 75 65
Sofia Persson, M.Sc., PhD	Post-doctoral fellow and molecular biologist/virologist at Swedish food safety agency Anna.Sofia.persson@slu.se +46 73 845 70 24
Sarah Coker, M.Sc., PhD	Research Engineer Leading laboratory work for Swedish environmental epidemiology center at SLU Cokersm6866@gmail.com +46 73 035 38 80